Acetylation increases

Writers:
EP300, KAT2A/B,
KAT5, KAT8, CREBBP

Erasers:
HDAC1-3,
SIRT1-3,6,7

Deacetylation decreases

Writer-dominant state (LHS)

Non-histone acetyltransferase activity

Splicing-factor lysine acetylation (43 sites; OR=7.36)

Blocks ubiquitination (competition)

Ub

Protein stabilization ($p = 0.62$)

SPLICEOSOME ACCUMULATION

107/111 proteins up-regulated · $p = 3.4 \times 10^{-37}$ · protein $p = +0.79$ · mRNA $p = -0.03$

U1 U4/U6 U5

Spliceosome complex
- snRNPs
- SF3B
- PRP19 complex
- NTC

Enhanced DNA repair

Ribosome biogenesis

Survival & proliferation

Immunosuppressive milieu

MIF

CD74

HDAC inhibitors

vorinostat • entinostat
HDAC2 docking – 7.9 kcal/mol

Spliceosome modulators

H3B-8800 • E7107
SF3B1 docking – 6.7 kcal/mol

Synergistic targeting